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CONTRIBUTION

Source: Ministry of Communications, India**Title:** Proposed baseline text for the proposed draft new ITU-T Technical Report JSTR.LCAP "Technical advances, challenges, and best practices in live captioning"**Contact:** Avinash Agarwal
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E-mail: rajivhkhattar@gmail.com**Abstract:** This contribution contains the draft baseline text for the proposed draft new ITU-T Technical Report JSTR.LCAP on Technical advances, challenges, and best practices in live captioning.**Summary and Rationale:** The objective of the Technical Report is to facilitate the implementation of live captioning and other visual accessibility features in Television Broadcasting, Cable Networks, Direct to Home (DTH), IPTV, other upcoming platforms, and emerging services such as Virtual Reality (VR). Member states will benefit from the experiences of other states, which the report will include as case studies.

The Technical Report will cover various aspects related to visual accessibility features, live captioning in particular, including recent advances, implementation challenges, and global best practices. The report will also discuss the captioning standards and accuracy measuring metrics, such as Word Error Rate (WER).

The Technical Report aims to provide a roadmap to make accessibility faster, accurate, and simpler using advanced technologies such as AI/ ML.

Proposal: The draft baseline text for the proposed draft new ITU-T Technical Report JSTR.LCAP is placed as ANNEX-1.

International Telecommunication Union

ITU-T Technical Report

TELECOMMUNICATION
STANDARDIZATION SECTOR
OF ITU

(06 SEP 2022)

JSTR.LCAP

**Technical advances, challenges, and best
practices in live captioning**

ITU-T

Summary

This Technical Report aims to provide guidance on technical aspects of live captioning and other visual accessibility features in Television Broadcasting, Cable Networks, Direct to Home (DTH), IPTV, other upcoming platforms, and emerging services such as Virtual Reality (VR) and also share global best practices through case studies.

The United Nations Convention on the Rights of Persons with Disabilities (UN CRPD) recognises that persons with disabilities also have similar and equal rights and the technology should enable the same. This document elaborates on the technological advancements and compliance framework to address this target group.

Keywords

Accessibility, captioning, WER, ASR, CRDP

Change Log

This document contains Version 1 of the ITU-T Technical Report JSTR.LCAPon "Technical advances, challenges, and best practices in live captioning" approved at the ITU-T Study Group 09 meeting held in Geneva, 06-14 Sept. 2022.

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Technical Report ITU-T JSTR.LCAP

ITU-T Technical Report on Technical advances, challenges, and best practices in live captioning

Summary

This Technical Report aims to provide guidance on technical aspects of live captioning and other visual accessibility features in Television Broadcasting, Cable Networks, Direct to Home (DTH), IPTV, other upcoming platforms, and emerging services such as Virtual Reality (VR) and also share global best practices through case studies.

The United Nations Convention on the Rights of Persons with Disabilities (UN CRPD) recognises that persons with disabilities also have similar and equal rights and the technology should enable the same. This document elaborates on the technological advancements and compliance framework to address this target group.

1 Scope

The objective of the Technical Report is to facilitate the implementation of live captioning and other visual accessibility features in Television Broadcasting, Cable Networks, Direct to Home (DTH), IPTV, other upcoming platforms, and emerging services such as Virtual Reality (VR). Member states will benefit from the experiences of other states, which the report will include as case studies.

The Technical Report covers various aspects related to visual accessibility features, live captioning in particular, including recent advances, implementation challenges, and global best practices. The report also discusses the captioning standards and accuracy measuring metrics, such as Word Error Rate (WER).

The Technical Report aims to provide a roadmap to make accessibility faster, accurate, and simpler using advanced technologies such as AI/ ML.

2 References

[1] To be added as the study progresses.

3 Terms and definitions

3.1 Terms defined elsewhere

This Technical Report uses the following terms defined elsewhere:

3.1.1 term [reference]: definition

3.1.2 term [reference]: definition

3.2 Terms defined here

This Technical Report defines the following terms:

3.1.1 term [reference]: definition

3.1.2 term [reference]: definition

[To be added as the study progresses.]

4 Abbreviations

ASR	Automatic Speech Recognition
WER	Word Error Rate

5 Introduction

In accordance with the United Nations Convention on the Rights of Persons with Disabilities (UN CRPD) that came into force in May 2008, persons with disabilities also have similar and equal rights and the technology should enable the same. This document elaborates on the technological advancements and compliance framework to address this target group.

Captioning is one of the most accepted accessibility enhancing techniques used in multimedia and television. Live captioning poses challenges in accurate and fast translation of the spoken words. Captioning standards are evolving and so are benchmarking practices.

The Technical Report aims to provide guidance on technical aspects and also share global best practices through case studies.

6 Captioning standards

[Editor's note: Text to be drafted – contributions encouraged.]

The Technical document will endeavor to study the current practices and will build a consensus to draft the standards , this will enable easy interfaces of the devices and implementation.

7 Quality and accuracy measurement

[Editor's note: Text to be drafted – contributions encouraged.]

7.1 WER

[Editor's note: Text to be drafted – contributions encouraged.]

7.2 NER

[Editor's note: Text to be drafted – contributions encouraged.]

8 Recent advances

[Editor's note: Text to be drafted – contributions encouraged.]

9 Use cases

[Editor's note: Text to be drafted – contributions encouraged.]

9.1 Education and Literacy

Access to the online educational content with captioning will go a long way in bringing the education to the doorsteps of the needy and will bring the disadvantaged students to the mainstream. Captioning can play an important role in disseminating education and promoting literacy even to the population that is not disadvantaged. Captioning enables people to learn languages faster and enhances reading capabilities. It is well established that any speech to text is remembered more and learnt faster as visual memories are retained much longer.

[Editor's note: Text to be drafted – contributions encouraged.]

9.2 Online Video Consumption

With the online video consumption on the rise and access to the handheld devices increasing, the captioning of online content can go a long way in bringing literacy and education to the remotest areas at ease of the viewers. As the online content moves from fixed-point viewing to on-choice viewing, the viewership will increase manifolds and captioning increases its accessibility to all. The defining of the standards and methodology will help in faster adoption.

[Editor's note: Text to be drafted – contributions encouraged.]

10 Case studies and global best practices

[Editor's note: Text to be drafted – contributions encouraged.]

11 Concluding Remarks

Live captioning is a vital tool for the industry to bring the benefits of technology to the most needed strata of society and will help in the nation-building. The industry must work together to bring in standards that help in reducing the implementation costs. The industry also needs to work with Regulators, licensors, Content owners and Distribution platforms to speed up the implementation.

[Editor's note: Text to be drafted – contributions encouraged.]

12 Bibliography

[To be added as the study progresses.]